

Fixed Points V: Curvature Coupling, Multi-Structure Dynamics, and Admissibility Barriers

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Version 9

Abstract

We formulate multi-structure covariance and admissibility diagnostics on the canonical FP–I–IV spine. Curvature enters through the full Laplace-type operator and its curved spectral projector, not through an unproved affine eigenvalue rule. Exact Gaussian claims are restricted to frozen linear Ornstein–Uhlenbeck systems. The scalar OU baseline is imported from Fixed Points III, its canonical owner, and the genuinely multi-structure stationary covariance derived here satisfies a Lyapunov equation, and block-diagonal damping gives a correct cross-covariance and canonical-correlation estimate. Admissibility is encoded by a finite or trace-controlled family of declared spectral, damping, and performance margins. The resulting deficit score detects exit but is neither a force nor an energy unless a separate variational source theorem is supplied. For affine Gaussian margin observables we prove finite-grid and continuous-time exit bounds. Cross-correlation controls simultaneous exits on a declared observation grid; it does not by itself prove localization, non-propagation, or selection of a post-exit state.

Version 9 Revision Note

Supersedes. Version 8, the previous authoring revision; version 7 remains the released edition.

Reason. Generic computational-evidence boilerplate implied that an external result repository supplied inputs to this analytic paper.

Resolution. The boilerplate is replaced by the actual dependency statement, and earlier FP sources are explicitly cited at their uses.

Retained result. All covariance, correlation and exit statements and proofs are unchanged.

Remaining boundary. Applications must specify the generator, noise, margins and any physical interpretation of the resulting estimates.

Version 8 Revision Note

Supersedes. Version 7, which remains the released edition. **Reason and resolution.** Separates the standalone analytic theorems from the physical source-selection problem.

Retained results. The previous mathematical results and proofs are unchanged. **Boundary.** This is a contextual clarification in an unreleased revision, not a new physical-source theorem.

Revision note for version 7

Supersedes. *Fixed Points V: Curvature Coupling, Multi-Structure Dynamics, and Admissibility Barriers*, version 6.

Reason. Version 6 corrected the covariance and exit mathematics, but a reader could still mistake the deficit diagnostic or cross-correlation for a force, propagation law, or post-exit selector.

Resolution. Version 7 adds a concrete covariance-cloud interpretation, an explicit explanation of why the deficit is diagnostic only, and a paper-specific route through the hypotheses. It preserves the scalar OU ownership of Fixed Points III and makes no new physical promotion.

Retained result. The stationary covariance, canonical-correlation, and declared-margin exit bounds are retained unchanged on their frozen-linear and Gaussian domains.

Remaining boundary. Nonlinear dynamics, causal propagation, localization, an exit-generating action, and selection of the next basin remain independent source problems.

Revision note for version 6

Supersedes. *Fixed Points V: Curvature Coupling, Multi-Structure Dynamics, and Drivers*, version 5.

Reason. Affine curvature shifts, nonlinear Gaussian claims, an incorrectly normalized covariance bound, and a deficit score interpreted as force, energy, censorship, and post-exit selection exceeded the proved model.

Resolution. Version 6 derives covariance from the frozen linear OU system, corrects canonical-correlation bounds, declares all admissibility margins, and proves only finite-grid or continuous Gaussian exit estimates. The scalar OU lemma is assigned to Fixed Points III and retained here only as a self-contained imported baseline.

Retained result. Multi-structure covariance and admissibility-exit diagnostics remain rigorous on their stated domains.

Remaining boundary. Correlation does not establish propagation, localization, an exit-causing force, or selection of the next basin.

How to read this paper

FP–III controls disturbances mode by mode, while FP–IV explains how curvature changes the operator and coherent projector. FP–V combines those inputs at a frozen configuration and asks two narrower questions: how are different modal structures statistically correlated, and how likely is a declared admissibility margin to be crossed?

The central picture in plain language. A stable linear stochastic system settles into a Gaussian cloud whose shape is the covariance matrix Σ . Its diagonal blocks describe uncertainty within individual structures; its off-diagonal blocks describe their shared fluctuation. Declared margins cut out an admissible region around the reference configuration. Exit estimates ask how much of the Gaussian cloud, or of a continuous Gaussian path, reaches the wrong side of those boundaries.

Why the deficit score is only diagnostic. The scalar $\mathfrak{B}$ records the worst violated margin, much as a single warning display can summarize many monitored constraints. It tells us whether the declared description has failed. It does not push the system toward or away from the boundary, assign an energetic cost, determine causal propagation, or choose the next basin. Any such role requires an additional source theorem connecting the score to an action or transition law.

Argument map. Sections 1–2 state the inherited curved linearization and import the scalar OU baseline from FP–III. Section 3 derives stationary covariance from the stable semigroup, then bounds cross-covariance and canonical correlation. Section 4 defines the admissibility margins and exact deficit diagnostic. Section 5 adds the affine Gaussian structure required for finite-grid and continuous-time exit bounds. Section 6 uses correlation to bound simultaneous observed exits and then states precisely what covariance cannot establish.

Scope boundary. The covariance theorems are frozen-linear results. The Gaussian exit theorems require affine margin observables. Nonlinear time-dependent operators, physical propagation, localization, and post-exit selection are not consequences of this layer.

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1 Inherited setting and scope

Let $I = [t_0, t_1]$ be a finite parameter interval and let (B, g_t) be a Riemannian base with uniformly bounded geometry on I . This formulation does not identify the FP stabilization parameter with physical Lorentzian time. Existence of a projected fixed point and promotion to a full equilibrium are inherited only when the hypotheses of FP I and FP II hold [2, 3]. Disturbance floors are interpreted according to FP III, and curvature uses the full operator and projector construction of FP IV [4].

A physical application must therefore select the coupled generator, projector, covariance source, invariant domain, and any post-exit transition kernel independently. The covariance

and exit theorems below diagnose those supplied data; they do not construct the application-specific physical operators.

At a frozen configuration x , write

$$L_x = \nabla_x^* \nabla_x + \mathcal{R}_x, \quad P_x = P_{\mathcal{R}}(x), \quad Q_x = I - P_x.$$

The contour defining P_x is assumed to remain in the resolvent set. If the old uncurved projector is used instead, the leakage term $Q\mathcal{R}P$ from FP IV must be retained.

All matrix theorems below are finite dimensional. They also extend to a separable Hilbert space when the noise covariance is trace class, the semigroup is exponentially stable, and every displayed covariance product is well defined. Modal sums are never used without a finite truncation or an explicit summability hypothesis.

2 Curved modal damping and exact OU modes

Let α denote a joint spectral label, including multiplicity. In a frozen Q_x -sector linearization, suppose the scalar mode satisfies the Itô equation

$$da_\alpha = -\gamma_\alpha(x)a_\alpha dt + \sqrt{q_\alpha(x)} dW_\alpha(t), \quad \gamma_\alpha(x) > 0. \quad (2.1)$$

Here γ_α is the damping margin obtained after the one-sided nonlinear estimate; it is not a shared deterministic/stochastic disturbance parameter. Curvature dependence is inherited from L_x . An affine response formula for an eigenvalue is an additional approximation, not a consequence of the Weitzenböck identity.

Imported scalar OU baseline. Fixed Points III is the canonical source for the exact scalar result [1]. Equation (2.1) has the unique invariant law

$$\mathcal{N}\left(0, \frac{q_\alpha}{2\gamma_\alpha}\right).$$

Deterministic forcing f_α instead gives the input-to-state scale $|f_\alpha|/\gamma_\alpha$; these two quantities are not interchangeable. For completeness, the variation-of-constants representation is

$$a_\alpha(t) = e^{-\gamma_\alpha t} a_\alpha(0) + \sqrt{q_\alpha} \int_0^t e^{-\gamma_\alpha(t-s)} dW_\alpha(s).$$

Itô isometry yields variance $q_\alpha(1 - e^{-2\gamma_\alpha t})/(2\gamma_\alpha)$, which converges to the displayed value. The deterministic scale follows by integrating the stable kernel against a bounded input.

3 Multi-structure covariance

Theorem 3.1 (Stable-semigroup covariance and resolvent bounds). *Consider*

$$d\zeta = A\zeta dt + B dW_t, \quad Q := BB^\top,$$

and assume

$$\|e^{tA}\| \leq M e^{-\omega t}, \quad t \geq 0, \quad M \geq 1, \quad \omega > 0.$$

Then the stationary covariance is

$$\Sigma = \int_0^\infty e^{tA} Q e^{tA^\top} dt,$$

it solves

$$A\Sigma + \Sigma A^\top + Q = 0,$$

and

$$\|\Sigma\| \leq \frac{M^2\|Q\|}{2\omega}, \quad \|A^{-1}\| \leq \frac{M}{\omega}.$$

Proof. The semigroup estimate makes the covariance integral converge. Differentiating its integrand gives the Lyapunov equation. Exponential stability also gives $A^{-1} = -\int_0^\infty e^{tA} dt$. Taking norms proves both bounds. $\square$

This form is required for nonnormal generators: the spectral abscissa alone does not control transient amplification or justify setting $M = 1$.

What the Lyapunov equation means. The covariance is accumulated noise transported by the decaying semigroup. The algebraic Lyapunov equation is the balance statement obtained after this accumulation has reached stationarity. The constant M remembers transient amplification of nonnormal dynamics; two generators with the same asymptotic decay rate can therefore support different covariance bounds.

For the block estimates and exit bounds below, specialize to the symmetric damping model and collect a finite set of frozen linearized amplitudes in $a(t) \in \mathbb{R}^d$:

$$da = -\Gamma a dt + B dW_t, \quad Q := BB^\top, \quad (3.1)$$

where $\Gamma = \Gamma^\top \succeq \gamma_0 I$ with $\gamma_0 > 0$.

Corollary 3.2 (Symmetric-damping covariance). *Equation (3.1) has a unique centered Gaussian invariant law with covariance*

$$\Sigma = \int_0^\infty e^{-t\Gamma} Q e^{-t\Gamma} dt.$$

It is the unique solution of

$$\Gamma\Sigma + \Sigma\Gamma = Q, \quad \|\Sigma\|_{\text{op}} \leq \frac{\|Q\|_{\text{op}}}{2\gamma_0}.$$

Proof. Apply Theorem 3.1 with $A = -\Gamma$, $M = 1$, and $\omega = \gamma_0$. Uniqueness follows by applying the stable semigroup to the homogeneous Lyapunov equation. $\square$

Partition the structures into A and B and assume for this section that

$$\Gamma = \text{diag}(\Gamma_A, \Gamma_B), \quad \Gamma_A \succeq \gamma_A I, \quad \Gamma_B \succeq \gamma_B I.$$

Then the cross block obeys the Sylvester equation

$$\Gamma_A \Sigma_{AB} + \Sigma_{AB} \Gamma_B = Q_{AB}.$$

Lemma 3.3 (Cross-covariance bound). *Under the block-damping hypothesis,*

$$\|\Sigma_{AB}\|_{\text{op}} \leq \frac{\|Q_{AB}\|_{\text{op}}}{\gamma_A + \gamma_B}.$$

In particular, $Q_{AB} = 0$ implies $\Sigma_{AB} = 0$ in this frozen linear model.

Proof. The Sylvester solution is $\Sigma_{AB} = \int_0^\infty e^{-t\Gamma_A} Q_{AB} e^{-t\Gamma_B} dt$. Taking norms proves the claim. $\square$

Definition 3.4 (Canonical cross-correlation). For a positive semidefinite covariance block matrix, define

$$\rho_{AB} := \|\Sigma_A^{\dagger/2} \Sigma_{AB} \Sigma_B^{\dagger/2}\|_{\text{op}}.$$

The covariance support inclusions imply $0 \leq \rho_{AB} \leq 1$. Let $m_A, m_B > 0$ denote the smallest positive eigenvalues of Σ_A, Σ_B on their supports.

Theorem 3.5 (Damping bound for canonical correlation). *Under the preceding hypotheses,*

$$\rho_{AB} \leq \frac{\|Q_{AB}\|_{\text{op}}}{(\gamma_A + \gamma_B)\sqrt{m_A m_B}}.$$

Consequently the right-hand side being smaller than $\rho_0 \in (0, 1)$ is a sufficient condition for $\rho_{AB} < \rho_0$.

Proof. Use submultiplicativity, the identities $\|\Sigma_A^{\dagger/2}\| = m_A^{-1/2}$ and $\|\Sigma_B^{\dagger/2}\| = m_B^{-1/2}$, and Lemma 3.3. $\square$

The smallest positive covariance eigenvalues are essential. Replacing them by $\|\Sigma_A\|$ and $\|\Sigma_B\|$ reverses the relevant normalization and does not bound canonical correlation.

Interpretation of cross-correlation. Q_{AB} is the shared noise input in the block model, while $\gamma_A + \gamma_B$ is the rate at which the two blocks jointly erase that shared fluctuation. Canonical correlation then normalizes the surviving cross-covariance by the variances actually present on each support. It is a dimensionless dependence measure, not a signal speed or causal kernel.

4 Admissibility margins and deficit score

Fix a finite family of continuous margins $m_j(x)$, $j \in \mathcal{J}$. They may include:

- separation of the curved low spectral cluster from the rest of the spectrum;
- positive one-sided damping margins $\gamma_\alpha(x)$;
- declared deterministic performance bounds $r_\alpha^{\text{det}} - |f_\alpha|/\gamma_\alpha$;
- declared stochastic variance bounds $r_\alpha^{\text{stoch}} - q_\alpha/(2\gamma_\alpha)$.

For an infinite family, replace finiteness by uniform convergence and compact control sufficient to make the infimum below continuous.

Definition 4.1 (Admissible domains and deficit score). Set

$$\mathcal{D}_\varepsilon := \{x : m_j(x) \geq \varepsilon \text{ for every } j \in \mathcal{J}\}, \quad \mathfrak{B}(x) := \max_{j \in \mathcal{J}} [-m_j(x)].$$

Then $x \in \mathcal{D}_0$ exactly when $\mathfrak{B}(x) \leq 0$, and $x \in \mathcal{D}_\varepsilon$ exactly when $\mathfrak{B}(x) \leq -\varepsilon$.

Proposition 4.2 (Diagnostic status). *The score $\mathfrak{B}$ is an exact scalar encoding of the declared margin constraints. It is not inserted into the evolution equation. It is not a physical energy, selection potential, or infinite enforcement cost unless an independent variational source theorem identifies it with such an object.*

Theorem 4.3 (Deterministic exit detection). *Let $X : I \rightarrow B$ be continuous and suppose the margins are continuous. If $X(t_0) \in \mathcal{D}_0$ and $\mathfrak{B}(X(t_1)) > 0$, there is a first exit time*

$$\tau := \inf\{t \in [t_0, t_1] : \mathfrak{B}(X(t)) > 0\},$$

with $\mathfrak{B}(X(\tau)) = 0$. At times with $\mathfrak{B}(X(t)) > 0$, at least one declared constraint fails.

Proof. The maximum of finitely many continuous functions is continuous. The result follows from the intermediate value theorem and the definition of first exit. $\square$

This theorem detects loss of the declared description. It neither supplies a force causing the exit nor selects the state or basin after exit.

What is exact here. For the chosen list of margins, $\mathfrak{B} \leq 0$ is exactly equivalent to admissibility. The theorem therefore provides an exact logical detector. The physical content still depends on whether the selected margins faithfully encode the intended geometry, damping, and performance requirements.

5 Gaussian exit estimates

We now impose the stronger structure actually needed for Gaussian claims. Assume the stationary solution of (3.1) and an exact affine linearized margin model

$$m_j(t) = \bar{m}_j + \ell_j^\top a(t), \quad \bar{m}_j > 0. \quad (5.1)$$

Thus $Z_j(t) := \ell_j^\top a(t)$ is a centered stationary Gaussian process with

$$v_j := \text{Var } Z_j(t) = \ell_j^\top \Sigma \ell_j \leq \|\ell_j\|^2 \frac{\|Q\|}{2\gamma_0}.$$

A merely Lipschitz nonlinear function of a Gaussian process is not generally Gaussian, so (5.1) cannot be replaced by Lipschitz regularity alone.

Theorem 5.1 (Finite-grid exit bound). *For observation times $t_1, \dots, t_N$,*

$$\mathbb{P}\left(\min_{j \in \mathcal{J}, 1 \leq k \leq N} m_j(t_k) < 0\right) \leq \sum_{j \in \mathcal{J}} N \exp\left(-\frac{\bar{m}_j^2}{2v_j}\right),$$

with a zero-variance term interpreted as zero when $\bar{m}_j > 0$.

Proof. For each pair (j, k) , the Gaussian tail bound gives $\mathbb{P}(Z_j(t_k) < -\bar{m}_j) \leq \exp[-\bar{m}_j^2/(2v_j)]$. Apply the union bound. No temporal independence is required. $\square$

Theorem 5.2 (Continuous-time Gaussian exit bound). *Assume each Z_j is separable with almost surely continuous paths and*

$$e_j := \mathbb{E} \sup_{t \in I} [-Z_j(t)] < \infty, \quad \sigma_j^2 := \sup_{t \in I} \text{Var } Z_j(t).$$

If $\bar{m}_j > e_j$ for every j , then

$$\mathbb{P}\left(\inf_{t \in I} \min_{j \in \mathcal{J}} m_j(t) < 0\right) \leq \sum_{j \in \mathcal{J}} \exp\left[-\frac{(\bar{m}_j - e_j)^2}{2\sigma_j^2}\right].$$

Proof. Apply the Borell–TIS inequality to the separable Gaussian process $-Z_j$ and then take a union bound over the finite constraint family. $\square$

Finiteness of e_j can be verified from the canonical increment metric by a metric-entropy bound. Exponential covariance decay alone suggests the usual $\sqrt{\log(1 + \gamma_0|I|)}$ scale, but that scale is not asserted without the required entropy estimate.

Finite grid versus continuous path. The grid theorem controls only the declared observation times and needs no temporal independence. The continuous theorem controls every time in the interval, but pays for that stronger statement through path continuity, separability, and a bound on the expected supremum. Sampling more densely does not by itself turn the first theorem into the second.

6 Simultaneous exits and the limit of correlation data

For two scalar affine margin observables at a fixed time, standardize the exit variables as $U = -Z_A/\sigma_A$ and $V = -Z_B/\sigma_B$. Suppose $-1 < \rho_0 < 1$, $\text{Corr}(U, V) \leq \rho_0$, and the standardized positive margins obey $u_A = \bar{m}_A/\sigma_A$, $u_B = \bar{m}_B/\sigma_B$.

Lemma 6.1 (Fixed-time simultaneous-exit bound).

$$\mathbb{P}(U \geq u_A, V \geq u_B) \leq \exp \left[-\frac{(u_A + u_B)^2}{4(1 + \rho_0)} \right].$$

For N declared observation times at which the same bounds hold, the probability of a simultaneous observed exit is at most N times the right-hand side.

Proof. The joint event implies $U + V \geq u_A + u_B$, while $\text{Var}(U + V) \leq 2(1 + \rho_0)$. Apply the one-dimensional Gaussian tail bound and then the union bound over observation times. $\square$

Proposition 6.2 (No propagation theorem from covariance alone). *Equal-time covariance or canonical correlation does not determine causal propagation, temporal clustering, or a post-exit transition. Such conclusions require a specified coupled evolution or transition kernel and control of cross-time covariance. Therefore FP V makes no probabilistic censorship or non-propagation claim from ρ_{AB} alone.*

7 Conclusion

FP V now supplies a rigorous frozen-linear covariance layer and a correctly scoped admissibility layer. Curvature is handled through the FP IV full operator and projector. Joint modes use the FP II/III indexing and disturbance conventions. Canonical correlation is normalized by the smallest positive covariance eigenvalues, and Gaussian persistence is proved only for affine Gaussian margin observables under the assumptions required by the relevant concentration theorem. The deficit score detects a declared loss of admissibility but is not promoted to dynamics, energy, propagation, or post-exit selection.

The practical chain is now explicit: a stable curved linearization determines a covariance; a declared affine margin turns that covariance into an exit observable; Gaussian concentration bounds its crossing probability; and cross-covariance can bound simultaneous observed crossings. The chain ends there. Moving from correlated exits to causal transport, localization, or a selected successor state requires a time-dependent coupled dynamics or transition kernel that is not encoded in equal-time covariance. That missing law is the next model-specific frontier, not another reinterpretation of the deficit score.

References

- [1] P. Nero, *Fixed Points III: Disturbance–Damping Balance and Stability*, version 8, 2026.

- [2] P. Nero, *Fixed Points I: Fixed Points over Multi-Bundle Manifolds*, version 7, 2026.
- [3] P. Nero, *Fixed Points II: Projected Fixed Points and Equilibria in a 10D Modal Model*, version 6, 2026.
- [4] P. Nero, *Fixed Points IV: Curvature, Centroid Motion, and Structural Transitions*, version 7, 2026.
- [5] G. Da Prato and J. Zabczyk, *Stochastic Equations in Infinite Dimensions*, Encyclopedia of Mathematics and Its Applications, Vol. 44, Cambridge University Press, Cambridge, 1992.
- [6] R. Temam, *Infinite-Dimensional Dynamical Systems in Mechanics and Physics*, Applied Mathematical Sciences, Vol. 68, Springer-Verlag, New York, 1997.

Dependency and reproducibility statement

All results use locally specified data, the preceding FP papers and standard mathematical literature. No later MTT paper or numerical packet supplies a premise. The covariance formulas and probability estimates can be evaluated once an application provides their inputs.